ASIA PACIFIC UNIVERSITY OF TECHNOLOGY AND INNOVATION

CT050-3-2-WAPP - WEB APPLICATIONS

GROUP ASSIGNMENT

LEARNING OUTCOMES

At the end of this assignment, you should be able to:

No.	Learning Outcome	Assessment Component
1	Develop a web application that is connected to a well-designed database by using server and client-side scripting language based on the current web standards. (C6, PLO2)	Group Assignment (Web-based System)
2	Explain the design and solution which provide a systematic way to develop web applications. (A3, PLO5)	Group Assignment (Documentation)

ASSIGNMENT BRIEF

As one of the members of a Web Application Development Team that consists of 4 or 5 students, you are assigned to design and develop a complete **Web-based Learning System using .Net technology**.

The objective is to create a website that will serve as a platform for registered and non-registered users looking for digital learning resources, which refer to digital resources such as web applications that engage students in learning activities and support students' learning goals. Online learning activities include but are not limited to synchronous and asynchronous online discussions, online self-assessments, virtual labs, simulations, problem-solving, and interactive learning objects.

The website must demonstrate database connectivity, client/ server processing, and good usage of multimedia elements. The team will entirely determine the website's title, scope, and contents. You are expected to have a local database, apply good interface design principles, and a fully functional navigation feature. The usage of a website template that suits the web application theme is allowed in this assignment.

REQUIREMENTS

The website must fulfill the following criteria, i.e., it should:

- Consist of interlinked webpages.
- Contain proper examples of appropriate usage of the HTML5 elements.
- Demonstrate Cascading Stylesheet (CSS) usages, this may be by providing:
  - External CSS
  - Internal CSS
  - Inline style
  - Combination of the above
- Provide good quality content.
- Demonstrate database connectivity to perform at least all the following operations:
  - Insert new records
  - Display records
  - Update/ modify records
  - Delete records
- Contain a registration page for new members.
- Contain registered member modules (require a valid login) to manage their activities.
- Contain administrator module (require a valid login) to manage the website content and database.
- Perform form validation.
- Provide appropriate navigation support.
- Perform proper file organization and naming convention.

Note: The usage of web page creation tools is permitted.

DOCUMENTATION

1. PROPOSAL REPORT

- Submission dateline: Week 7

Groups are required to submit a proposal report which consists of the following:

- Website Title
- Objectives
- Mission Statement
- Audience Modelling
- Project Scope

2. FINAL REPORT

The documentation should include the following:

- Format:
  - Type face: Times New Roman. Boldface, italics & lines can be used for emphasizing and to enhance readability.
  - Font size: 12(except titles and headings).
  - Spacing: 1.5 lines between texts of a paragraph.
  - Alignment: justify.
  - Headers and footers can be used.
  - Numbered.
  - Binding: APU standard for Level 2
  - Cover page, your cover page should contain the following items
    - » Group Member's Name and ID.
    - » Intake code.
    - » Subject.
    - » Project Title.
    - » Date Assigned (the date the report was handed out).
    - » Date Completed (the date the report is due to be handed in).
    - » APU logo

- Recommended Content
  - Table of Contents
    - » The table of contents should have the topic title and reference page number attached to each topic.
  - Introduction/ Project Plan
    - » This section introduces your project including the objectives, scopes, and project schedule.
  - Requirement Specification
    - » This section defines the audience modeling, use cases, flow charts, and major functions of the web.

- Design and Modeling
  - » Provide Entity Relationship Diagram (ERD), Wireframes for major layouts of the web pages, and Website Navigational Structure. All components must be provided along with appropriate descriptions.
- Implementation
  - » Provide a detailed explanation of the source code of the major web application features. This includes CSS for web page styling, form validation, and SQL queries for database connectivity.
  - »
- User Guidance
  - » Screenshots of the web pages with an appropriate description.
- Conclusions (summary, lesson learned, and future enhancements)
- References
- Appendix
  - » Proposal Report

- **Note: Full source code should not be included in the documentation**.

FINAL REPORT SUBMISSION

- The complete Web site and report are to be submitted in Moodle.

Marking Schema

Group Assignment - Documentation (50%):

No	Documentation Group Criteria	Mark Allocated (%)	Score (%)	Remark
1	Introduction	10	0	
2	Storyboard & Requirement Specification	10	0	
3	Design & Modelling	10	0	
4	Implementation & Discussion	10	0	
5	Conclusion, Document Styles & Formatting	10	0	
	Total:	**50**	**0**	

Group Assignment - Implementation (50%):

No	Implementation (Website and Presentation) Individual Criteria	Mark Allocated (%)	Score (%)	Remark
1	Web Page Layout & Appearance	10	0	
2	User Authentication & Authorization	10	0	
3	Dynamic Content	10	0	
4	Insert, Update & Delete Records	10	0	
5	Form Validation, Navigation & Usability	10	0	
	Total:	**50**	**0**	

MARKING RUBRIC TABLE

REPORT

Criteria	9–10 (Distinction)	7–8 (Credit)	5–6 (Pass)	4 (Marginal Fail)	0–3 (Fail)
Introduction	Comprehensive and engaging introduction, well-articulated goals and objectives.	Well-defined problem statement with clear goals and objectives.	Basic introduction with some mention of goals and objectives, but lacks depth or clarity.	Weak introduction with unclear problem statement, goals, and objectives; lacks clarity and detail.	Missing or unclear introduction, vague problem statement, no clear goal or objectives.
Storyboard and Requirement Specification	Comprehensive storyboard with clear user flow; requirements are precise, detailed, and well-documented.	Well-structured storyboard with clear flow; detailed requirements covering major system aspects.	Basic storyboard and requirements provided but lack clarity, detail, or completeness.	Limited storyboard with unclear flow; requirements are incomplete with missing key details.	Missing or poorly structured storyboard; requirements are unclear or not defined.
Design & Modelling	Fully developed, accurate, and efficient design with detailed modeling.	Well-structured design with appropriate models and clarity.	Basic design, but lacks detail or alignment with requirements.	Basic design with weak structure, lacking clarity, detail, and alignment with requirements.	No or minimal design, lacking logical structure.
Implementation & Discussion	Fully implemented system with thorough and critical discussion.	Mostly functional implementation with insightful discussion.	Partially implemented system with basic discussion.	Incomplete implementation with limited functionality and minimal or unclear discussion.	Incomplete implementation with little to no discussion.
Conclusion, Document Styles & Formatting	High-quality formatting and well-structured, insightful conclusion.	Clear and professional formatting, effective conclusion.	Basic conclusion and formatting with some inconsistencies.	Weak conclusion with inconsistent formatting and limited clarity or structure.	Missing or poorly formatted document, weak conclusion.

WEBSITE

Criteria	9–10 (Distinction)	7–8 (Credit)	5–6 (Pass)	4 (Marginal Fail)	0–3 (Fail)
Web Page Layout & Appearance	Outstanding layout with excellent design and usability considerations.	Good design, visually appealing, and properly structured layout.	Functional layout but lacks aesthetic appeal or consistency.	Weak layout with inconsistent design and limited visual appeal.	Poor visual appeal and layout with inconsistent design.
User Authentication & Authorization	Robust authentication and authorization with high-level security.	Secure authentication with appropriate user roles.	Basic authentication but lacks security measures.	Weak authentication with limited security and unclear access control.	Missing or weak authentication mechanism.
Dynamic Content	Advanced, highly interactive dynamic content implementation.	Well-integrated dynamic content with smooth functionality.	Limited dynamic interactions but not fully functional.	Limited dynamic content with partial or inconsistent functionality.	Static content with no dynamic elements.
Insert, Update & Delete Records	Optimized, secure, and well-integrated database operations.	Functional insert, update, and delete operations with validations.	Basic CRUD operations but lacking efficiency or security.	Basic CRUD operations with incomplete functionality or issues.	No or incorrect database operations.
Form Validation, Navigation & Usability	Seamless usability, robust validation, and intuitive navigation.	Well-implemented validation and logical navigation.	Basic validation but lacks user-friendly navigation.	Basic validation with weak navigation and limited usability.	Poor form validation and difficult navigation.

10.0 AI USAGE POLICY (TRAFFIC LIGHT SYSTEM)

🟥 RED – STRICTLY PROHIBITED

Students **must NOT use AI tools** to generate or modify:

- Any **code implementation** (client-side or server-side)
- **Program logic**, algorithms, or workflows
- **Database queries** (e.g., SQL statements)
- **System design**, architecture, or OOP structures
- Debugging that results in **AI providing corrected code**

  Rule: If AI produces code or logic that appears in your submission

  → it is considered misconduct.

🟧 ORANGE – RESTRICTED USE / LIMITED USE

AI may be used **only for language support**, not content creation:

- Grammar and spelling correction
- Improving clarity and readability of writing
- Formatting references (APA/IEEE, etc.)

NOT ALLOWED:

- Generating explanations, descriptions, or report content
- Paraphrasing AI-generated technical explanations

Rule: All submitted content must reflect your own understanding and writing.

🟩 GREEN – ALLOWED

AI tools may be used for **learning purposes only**, not for direct submission:

- General learning support
- Syntax debugging only
- Identifying **error types** (e.g., “null reference error meaning”)

NOT ALLOWED

- Copying AI-generated examples into your assignment
- Asking AI to solve your assignment tasks directly

Rule: You may learn from AI, but **you must independently produce your final work**.

Declaration Requirement

Students may be required to declare and justify AI usage. Failure to declare AI usage may be treated as academic misconduct.

AI Usage Declaration Form

Group Members:
Student ID: _____ Student Name: _____
Student ID: _____ Student Name: _____
Student ID: _____ Student Name: _____
Student ID: _____ Student Name: _____

Section A: AI Usage Disclosure

Have you used any AI tools (e.g., ChatGPT, Grammarly, Copilot, etc.)?
☐ Yes ☐ No

Section B: Details of AI Usage (Complete ONLY if Yes)

1. List the AI Tool(s) Used:

_____
_____

2. Purpose of Use (Tick all that apply):
☐ Grammar checking
☐ Improving writing clarity
☐ Formatting references
☐ Understanding concepts
☐ Debugging syntax errors
☐ Other: _____
3. Description of how AI was used:

_____
_____
_____
_____
_____

4. Sections where AI was applied:
☐ Report writing
☐ References
☐ Documentation formatting
☐ Other: _____

Section C: Academic Integrity Declaration

We confirm that the submitted work is our own original work, AI tools were used only within allowed scope, and no prohibited use was made.

Name	Signature	Date